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Health monitoring system and device for livestock

Abstract

The present disclosure generally relates to a monitoring device (**100**) for remotely monitoring health of a young livestock animal (**10**), the device (**100**) comprising: an elongated body (**110**) within predefined dimensional limits; and a set of electronic components housed within the elongated body (**110**), the electronic components comprising: a temperature sensor for measuring body temperature of the young livestock animal (**10**); a computer processor for processing the measured temperature into temperature data; and a wireless communications component for transmitting the temperature data to a remote computer device (**210**) for processing to monitor the health of the young livestock animal (**10**), wherein the predefined dimensional limits of the elongated body (**110**) enable the monitoring device (**100**) to be orally administrable into the young livestock animal (**10**) and to remain inside its stomach (**12**) without being excreted.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to a health monitoring system and device for livestock. More particularly, the present disclosure describes various embodiments of a monitoring device for remotely monitoring health of a livestock animal, as well as a system for monitoring the health of livestock animals using the monitoring devices.

BACKGROUND

(2) Livestock are reared in farms for various purposes including producing food for human consumption. A farm's productivity from livestock is dependent on their health and it is important to be able to monitoring their health constantly. Body temperature monitoring is one of the most effective ways of evaluating the health of livestock animals to ensure they are not compromised by infection or disease. Farmers traditionally use thermometers to measure body temperature of each of their livestock animals rectally. However, this labour-intensive manual measuring of body temperature on the farm is very tedious and inefficient.

(3) Therefore, in order to address or alleviate at least one of the aforementioned problems and/or disadvantages, there is a need to provide an improved monitoring system and device for livestock.

SUMMARY

(4) According to a first aspect of the present disclosure, there is a monitoring device for remotely monitoring health of a young livestock animal, the device comprising: an elongated body within predefined dimensional limits; and a set of electronic components housed within the elongated body, the electronic components comprising: a temperature sensor for measuring body temperature of the young livestock animal; a computer processor for processing the measured temperature into temperature data; and a wireless communications component for transmitting the temperature data to a remote computer device for processing to monitor the health of the young livestock animal, wherein the predefined dimensional limits of the elongated body enable the monitoring device to be orally administrable into the young livestock animal and to remain inside its stomach without being excreted.

(5) According to a second aspect of the present disclosure, there is a monitoring system for remotely monitoring health of livestock animals comprising young livestock animals, the

monitoring system comprising: a set of monitoring devices; a remote computer device communicative with the monitoring devices; and each monitoring device residing in a stomach of a respective livestock animal, the monitoring device comprising: an elongated body within predefined dimensional limits that enable the monitoring device to remain inside its stomach without being excreted and to be orally administered into the stomach of a young livestock animal; a set of electronic components housed within the body, the electronic components comprising: a temperature sensor for measuring temperature of the livestock animal; a computer processor for processing the measured temperatures into temperature data; and a wireless communications component for transmitting the temperature data to the remote computer device, wherein the remote computer device is configured for processing the temperature data to monitor the health of the livestock animals.

(6) A monitoring system and device for livestock according to the present disclosure are thus disclosed herein. Various features, aspects, and advantages of the present disclosure will become more apparent from the following detailed description of the embodiments of the present disclosure, by way of non-limiting examples only, along with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of an exterior of a monitoring device according to embodiments of the present disclosure.
- (2) FIG. 2 is an illustration of an interior of the monitoring device.
- (3) FIG. 3 is an illustration of a livestock animal having the monitoring device residing inside its stomach.
- (4) FIG. 4 is an illustration of the monitoring device comprising a data matrix code.
- (5) FIG. 5 is an illustration of the monitoring device being activated using an external magnetic device.
- (6) FIG. 6 is an illustration of a rumen bolus applicator for inserting the monitoring device into the livestock animal.
- (7) FIG. 7 is an illustration of a monitoring system comprising a set of the monitoring devices.

DETAILED DESCRIPTION

- (8) For purposes of brevity and clarity, descriptions of embodiments of the present disclosure are directed to a monitoring system and device for livestock, in accordance with the drawings. While aspects of the present disclosure will be described in conjunction with the embodiments provided herein, it will be understood that they are not intended to limit the present disclosure to these embodiments. On the contrary, the present disclosure is intended to cover alternatives, modifications and equivalents to the embodiments described herein, which are included within the scope of the present disclosure as defined by the appended claims. Furthermore, in the following detailed description, specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be recognised by an individual having ordinary skill in the art, i.e. a skilled person, that the present disclosure may be practiced without specific details, and/or with multiple details arising from combinations of aspects of particular embodiments. In a number of instances, well-known systems, methods, procedures, and components have not been described in detail so as to not unnecessarily obscure aspects of the embodiments of the present disclosure.
- (9) In embodiments of the present disclosure, depiction of a given element or consideration or use of a particular element number in a particular figure or a reference thereto in corresponding descriptive material can encompass the same, an equivalent, or an analogous element or element number identified in another figure or descriptive material associated therewith.
- (10) References to “an embodiment/example”, “another embodiment/example”, “some

embodiments/examples”, “some other embodiments/examples”, and so on, indicate that the embodiment(s)/example(s) so described may include a particular feature, structure, characteristic, property, element, or limitation, but that not every embodiment/example necessarily includes that particular feature, structure, characteristic, property, element or limitation. Furthermore, repeated use of the phrase “in an embodiment/example” or “in another embodiment/example” does not necessarily refer to the same embodiment/example.

(11) The terms “comprising”, “including”, “having”, and the like do not exclude the presence of other features/elements/steps than those listed in an embodiment. Recitation of certain features/elements/steps in mutually different embodiments does not indicate that a combination of these features/elements/steps cannot be used in an embodiment.

(12) As used herein, the terms “a” and “an” are defined as one or more than one. The use of “/” in a figure or associated text is understood to mean “and/or” unless otherwise indicated. The term “set” is defined as a non-empty finite organisation of elements that mathematically exhibits a cardinality of at least one (e.g. a set as defined herein can correspond to a unit, singlet, or single-element set, or a multiple-element set), in accordance with known mathematical definitions.

(13) The present disclosure describes remote health monitoring of livestock. Livestock may be defined as domesticated animals raised in an agricultural setting to produce labour and commodities, such as food (e.g. meat, eggs, and milk) and materials (e.g. fur and wool). The United States defines the term “livestock” as including cattle, sheep, horses, goats, and other domestic animals ordinarily raised or used on the farm. This term can sometimes be used to refer solely to animals that are bred for consumption, while other times it may refer only to farmed ruminants, such as cattle and goats. Ruminants may be defined as herbivorous mammals that are able to acquire nutrients from food by fermenting it in a specialized stomach prior to digestion. The ruminant stomach is a multi-chambered organ and usually has four separate chambers—rumen, reticulum, omasum, and abomasum. The rumen and reticulum can be collectively referred to as the reticulorumen.

(14) In representative or exemplary embodiments of the present disclosure, with reference to FIG. 1 and FIG. 2, there is a monitoring device **100** for remotely monitoring health of a livestock animal. The livestock animal can be an adult or a young and can be a ruminant. In many embodiments, the monitoring device **100** is used for remotely monitoring health of a young livestock animal, such as a young ruminant like a calf.

(15) The monitoring device **100** includes an elongated body **110** that is within predefined dimensional limits. The predefined dimensional limits enable the monitoring device **100** to be orally administrable into the young livestock animal and to remain inside its stomach without being excreted. More specifically, the monitoring device **100** is small enough so that the monitoring device **100** can be orally administered, i.e. from the mouth through the oesophagus and into the stomach of the young livestock animal which is smaller than an adult one. If the monitoring device **100** is small enough for the young livestock animal, it would be orally administrable into an adult livestock animal. Further, the monitoring device **100** is large enough so that the monitoring device **100** stays inside its stomach and does not get excreted naturally. This allows the monitoring device **100** to be used for long term monitoring of the health of the young livestock animal.

(16) FIG. 3 illustrates an example of a livestock animal **10** having the monitoring device **100** residing in its stomach **12**. The livestock animal **10** may be a ruminant, such as an adult cattle or young calf, and the monitoring device **100** resides in its reticulorumen. The monitoring device **100** may be referred to as a rumen bolus as it can reside in the rumen.

(17) Based on the predefined dimensional limits, the elongated body **110** may have an overall width of up to 25 ± 0.5 mm, and preferably up to 24 ± 0.5 mm. Lengthwise, the elongated body **110** may have an overall length of up to 100 ± 0.5 mm. The elongated body **110** may be tubular with an overall diameter of up to 25 ± 0.5 mm, and preferably up to 24 ± 0.5 mm. In some embodiments as shown in FIG. 1, the elongated body **110** includes a rounded front portion **112** to facilitate oral

administration of the monitoring device **100** into the mouth of the livestock animal **10**.

(18) The elongated body **110** is made of a material that is non-toxic and safe for consumption as the monitoring device **100** would be residing in the stomach **12**. The material is also resistant to aggressive/corrosive elements that are present in the stomach **12**, such as acids and microbes that aid digestion, so that the monitoring device **100** can stay inside the stomach **12** without becoming severely damaged or being excreted out of the stomach **12** for a period of approximately up to 4 years.

(19) As an example, the elongated body **110** may include a thermal epoxy material. As the monitoring device **100** is intended to remain entrenched in the stomach **12** for approximately up to 4 years, the electronic components are encapsulated in the elongated body **110** made of a special thermal epoxy material that protects the electronic components from being oxidized from heat generated in the stomach **12** and liquids in the stomach **12**. Additionally, the elongated body **110** may be coated with a suitable material that is exposed to liquids in the stomach **12**. Some non-limiting examples of the material are plastics approved by the United States Food and Drug Administration, such as food grade polypropylene. The polypropylene coating makes the monitoring device **100** safe to remain entrenched in the stomach **12** and the underlying thermal epoxy material protects electronic components from substances such as water vapour that may have permeated through the polypropylene coating over a long period of time.

(20) The monitoring device **100** may include a ballast weight **114** housed within the elongated body **110** to increase the overall weight and density of the monitoring device **100**. The ballast weight **114** may be made of a metallic material. The ballast weight **114** may be positioned closer to the front portion **112**, so that the centre of mass of the monitoring device **100** is shifted towards the front portion **112**. This arrangement front weights the monitoring device **100** to facilitate the oral administration. More specifically, the front weighting assists in directing travel of the monitoring device **100** through the oesophagus of the livestock animal **10**, including a young one.

(21) In one working prototype of the monitoring device **100**, the elongated body **110** has a tubular profile with a rounded front portion **112**, an overall length of 100 mm, and overall width of 25 mm. In another working prototype, the elongated body **110** has an overall length of 100 mm, and overall width of 24 mm.

(22) The monitoring device **100** has an overall weight from approximately 85 to 95 g, preferably from approximately 87 to 89 g, and more preferably at approximately 88 g. Based on the dimensions of the monitoring device **100** as described above, the monitoring device **100** has an overall density from approximately 2.0 to 2.1 g/cm.³, preferably from approximately 2.03 to 2.08 g/cm.³, and more preferably at approximately 2.03 g/cm.³.

(23) The overall size, weight, and density of the monitoring device **100**, together with the front weighting of the monitoring device **100**, enable the monitoring device **100** to remain lodged in the stomach **12** without being excreted out (for up to 4 years) and to ameliorate discomfort to the livestock animal **10**. Moreover, at the preferred overall weight and density, the monitoring device **100** would be in a floated position inside the stomach **12**, enabling higher gain for antenna reception (by a wireless communications component of the monitoring device **100**), and minimizing unnecessary disruption for transmission of temperature data, as described further below. The front weighting by the ballast weight **114** lowers the centre of mass of the monitoring device **100** when it is lodged in the stomach **12**, enabling the monitoring device **100** to float in a more upright position for optimal wireless communications.

(24) The monitoring device **100** further includes a set of electronic components housed within the elongated body **110**. More specifically, the electronic components are encased inside the elongated body **110** such that the monitoring device **100** is tamperproof and the electronic components are not exposed to external elements. The elongated body **110** forms a shell casing over the electronic components and protects them against the aggressive/corrosive elements inside the animal's stomach **12**. Preferably, the enclosure protection provided by the elongated body **110** meets the

Ingress Protection Code 67 standard.

(25) The electronic components include a temperature sensor for measuring body temperature of the livestock animal **10**. More specifically, as the monitoring device **100** will reside in the animal's stomach **12**, the temperature sensor measures, within an accuracy of $\pm 0.2^{\circ}$ C., the internal body temperature of the stomach **12**. It will be appreciated that various types of temperature sensor can be used, as will be readily understood by the skilled person. The electronic components may include other sensors to measure other physiological parameters of the livestock animal **10**. Some non-limiting examples of such other sensors are pH sensors to measure the pH values (i.e. acidity/alkalinity) of liquids inside the stomach **12**. The pH values can be used to identify the onset flatulence or bloating in the stomach **12** so that such conditions can be quickly remedied to improve the health of the livestock animal **10**.

(26) The electronic components further include a computer processor for processing the measured temperature into temperature data, and a wireless communications component for transmitting the temperature data to a remote computer device. More specifically, the temperature sensor measures the raw temperature values of the animal's internal body temperature. The computer processor receives the temperature values from the temperature sensor and processes them into electronic temperature data for onward transmission by the wireless communications component. As the monitoring device **100** is inside the animal's stomach **12**, the wireless communications component transmits the temperature data using suitable wireless communication protocols, preferably for long range communications, as will be described further below. The remote computer device is configured for receiving and processing the temperature data to monitor the health of the livestock animals **10**.

(27) The computer processor may be configured for controlling the temperature sensor and wireless communications component to perform measuring of the body temperature at predefined intervals and transmitting the temperature data at predefined intervals. For example, the temperature sensor is configured to measure the body temperature at 15-minute intervals, while the wireless communications component is configured to transmit the temperature data at 30-minute intervals. The computer processor may be configured to adjust the measurement and transmission intervals as desired, such as based on instructions received from the remote computer device.

(28) The temperature sensor, computer processor, and wireless communications component may be integrated together in a printed circuit board assembly (PCBA) **120**. The electronic components may further include an antenna component **122** for boosting the wireless transmissions to the remote computer device. The antenna component **122** may be integrated in the PCBA **120** or separately connected to the PCBA **120**.

(29) The electronic components further include a battery **124** for powering the electronic components. The battery **124** is separately connected to the PCBA **120** and is designed with a unique battery management capability such that the battery **124** has a battery lifespan of approximately up to 4 years. This means the monitoring device **100** can continue measuring the body temperature and transmit the temperature data for approximately up to 4 years. Additionally, the elongated body **110** encasing the battery **124** also seals any chemical leakages from the **124** within the elongate body **110**, thereby protecting the livestock animal **10** from such chemical leakages.

(30) In some embodiments, the electronic components optionally include a data storage or memory component connected to the computer processor for storing the temperature data. The memory component has a capacity to store a predefined number of temperature data. For example, the memory component can store 16 temperature values measured at 15-minute intervals, equivalent to approximately 4 hours of temperature data. The memory component serves as a backup storage in case the monitoring device **100** is out of range of the remote computer device, particularly if the livestock animal **10** has wandered too far away. The temperature data that has not been transmitted, as well as new temperature data that has been processed during the period when the monitoring

device **100** was out of range, would be transmitted to the remote computer device when the monitoring device **100** returns to communication range.

(31) In some embodiments, the electronic components optionally include an electronic identification device to enable electronic identification of the particular livestock. For example, the identification device is a short range identification device or transponder such as a RFID (radio frequency identification) device or a NFC (near field communication) device. The livestock animal **10** having the monitoring device **100** residing inside its stomach **12** may be identified when it passes through a gantry or gate reader that detects the identification device. Regulatory bodies in many European countries mandate the use of RFID devices or transponders for identification of livestock animals **10**. By incorporating the RFID device or transponder in the monitoring device **100** residing in the animal's stomach **12**, the RFID device or transponder can be detected when the livestock animal **10** passes through the gantry or gate reader, thereby identifying and registering the livestock animal **10**.

(32) To enable wireless communications between the monitoring device **100** and the remote computer device, the monitoring device **100** first needs to be paired with the remote computer device. The monitoring device **100** is pre-programmed or predefined with a set of identification data or credentials, such as including a network address for use in communications within a network centred at the remote computer device. One example of the network address is a media access control (MAC) address.

(33) Depending on the wireless communication protocol, the predefined identification data or device credentials may include those that are unique to the protocol, such as a low-power wide-area network protocol. An example of such protocol is LoRa which may employ the LoRaWAN® network architecture and specification. The LoRaWAN® specification describes certain device credentials that should be predefined for the monitoring device **100**. The device credentials in the LoRaWAN® specification may include DevEUI, JoinEUI, and AppKey, as will be readily understood by the skilled person.

(34) In one embodiment, the predefined identification data for the monitoring device **100** is provided by the manufacturer in printed form, such as in the operations manual. In another embodiment, the manufacturer provides a link to a secured website for retrieval of the predefined identification data. In another embodiment and as shown in FIG. 4, the monitoring device **100** includes a data matrix code **130** for retrieving the predefined identification data. The data matrix code **130** may be in the form of a QR code disposed at a rear portion **116** of the elongated body **110**. The data matrix code **130** may be integrally formed or embedded, such as by direct printing, on the rear portion **116**. Alternatively, the data matrix code **130** is visible on a label or sticker, such as by printing thereon, that is detachably attached to the rear portion **116**.

(35) The predefined identification data can be retrieved by using an electronic device **140**, such as a mobile phone or tablet device having a camera, to scan the data matrix code **130**. In one example, the predefined identification data is encoded in the data matrix code **130** and retrieved in response in response to the scan. In another example, an online address link is encoded in the data matrix code **130** and the electronic device **140** is directed to the online address, such as the secured website mentioned above, for retrieval of the predefined identification data. The retrieved predefined identification data or device credentials may subsequently be used for pairing and registration of the monitoring device **100** with the remote computer device. For example, the device credentials are entered into the remote computer device for it to identify and pair with the monitoring device **100**.

(36) The monitoring device **100** is initially in an inactive or standby mode wherein the power consumption from the battery **124** is minimal. In the inactive mode, the monitoring device **100** does not measure any temperature nor transmit any temperature data. Before orally administering the monitoring device **100** into the livestock animal **10**, the monitoring device **100** needs to be activated to the active mode.

(37) The monitoring device **100** may include a magnetic mechanism **132** housed within the elongated body **110** and configured for activating the monitoring device **100**. Like the electronic components, the magnetic mechanism **132** is encased inside and shielded by the elongated body **110**, and is operative by a magnetic field. More specifically, as shown in FIG. 5, an external magnetic device **142** is placed near the magnetic mechanism **132**. The external magnetic device **142** generates a magnetic field and causes magnetic interference **134** with the magnetic mechanism **132**. The monitoring device **100** is activated in response to the magnetic mechanism **132** detecting the magnetic interference **134**. A detachable label or sticker may be placed on the surface of the elongated body **110** to locate the magnetic mechanism **132** housed therein.

(38) As shown in FIG. 5, the monitoring device **100** may include a lighting element **136**, such as including a light emitting diode (LED) housed within the elongated body **110**. Like the electronic components, the magnetic mechanism **132** is encased inside and shielded by the elongated body **110**. The elongated body **110** may include a transparent/translucent area so that light signals generated by the inside lighting element **136** can be seen. The lighting element **136** is configured for visual indication of activation of the monitoring device **100**. For example, the lighting element **136** may generate a light signal (such as by flashing a number of times, e.g. 5) in response to successful activation of the monitoring device **100**. The lighting element **136** may generate a different light signal if the activation fails.

(39) Upon activation of the monitoring device **100**, the monitoring device **100** attempts to join or register with the remote computer device and transmits an activation data packet to the remote computer device. The activation data packet includes the predefined identification data or device credentials of the monitoring device **100**. The remote computer device receives the activation data packet and matches the device credentials. If the device credentials match those that has been entered into the remote computer device, the monitoring device **100** is successfully paired and registered with the remote computer device. The lighting element **136** may generate another light signal (such as by flashing a number of times, e.g. 5) in response to successful registration of the monitoring device **100** with the remote computer device. The flashes in the light signals for successful activation and successful registration may be at different frequencies to differentiate the light signals. The lighting element **136** may generate a different light signal if the registration fails.

(40) The registration may fail due to various possible reasons, such as if the device credentials were incorrectly entered into the remote computer device or if the monitoring device **100** is out of range. The monitoring device **100** may be configured for a reset to the inactive mode before reattempting registration with the remote computer device.

(41) In one embodiment, the reset can be performed by placing the external magnetic device **142** near the magnetic mechanism **132**. The reset is initiated in response to the magnetic mechanism **132** detecting the magnetic interference **134**. The lighting element **136** may generate a light signal (such as by flashing a number of times, e.g. 3) in response to successful reset of the monitoring device **100**. The monitoring device **100** reverts to the inactive mode after the successful reset. Repeating the steps described above, the external magnetic device **142** can be used to activate the monitoring device **100** and reattempt registration with the remote computer device.

(42) Upon successful registration of the monitoring device **100** with the remote computer device, wireless communications between the monitoring device **100** and the remote computer device is enabled. The monitoring device **100** begins to measure temperature and transmit temperature data to the remote computer device according to the predefined intervals. The lighting element **136** may generate a light signal (such as by flashing once) every time the temperature data is transmitted. Additionally, the monitoring device **100** may be configured to disable the magnetic mechanism **132** so that the monitoring device **100** cannot be reset anymore. This would mean that the monitoring device **100** is permanently paired with the remote computer device.

(43) After successfully activation and registration of the monitoring device **100**, the monitoring device **100** needs to be orally administered into the livestock animal **10**. For example, as shown in

FIG. 6, a rumen bolus applicator **150** can be used to insert the monitoring device **100**, which is in the form of a rumen bolus, into the livestock animal **10**. As described above, the monitoring device **100** is sized to fit in the oesophagus of the livestock animal **10**, including a young ruminant, and sit in the stomach **12** (e.g. reticulorumen) without being excreted.

(44) In some embodiments, the monitoring device **100** has detachable labels or stickers on the elongated body **110**, such as to show the data matrix code **130** and locate the magnetic mechanism **132**. These labels should be detached from the monitoring device **100** before orally administering the monitoring device **100**, as the labels may contain contaminants potentially harmful to the livestock animal **10**.

(45) The monitoring device **100** may include a human-readable identifier integrally formed on the elongated body **110**. This human-readable identifier provides a means for identifying the monitoring device **100** among a large number of monitoring devices **100**, since a farm has a large number of livestock animals **10**. The human-readable identifier may be a string of text characters, such as numbers that number the monitoring devices **100** sequentially. The human-readable identifier may be integrally formed on any part of the elongated body **110**, such as on the rear portion **116** so that the monitoring device **100** can be identified after the label containing the data matrix code **130** is removed. The human-readable identifier may be integrally formed using various means such as laser printing and engraving.

(46) In representative or exemplary embodiments of the present disclosure, with reference to FIG. 7, there is a monitoring system **200** for remotely monitoring health of livestock animals **10** including young livestock animals. The monitoring system **200** includes a set of one or more monitoring devices **100** and the remote computer device communicative with the monitoring devices **100**. Each monitoring device **100** resides in the stomach **12** of a respective livestock animal **10** and has the elongated body **110** within the predefined dimensional limits such that the monitoring device **100** is orally administered into the livestock animal **10** (even a young one) and remains inside the stomach **12** without being excreted.

(47) In many embodiments, the communication between the monitoring devices **100** and the remote computer device is based on the LoRa wireless communication protocol which employs the LoRaWAN® specification. The LoRaWAN® protocol enables data transmissions at low power and over long distances up to a few kilometres, making it suitable for use in livestock farms that span across large areas of land. Additionally, the LoRaWAN® protocol has a large network capacity for the network of monitoring devices **100**, making it suitable for use with large numbers of livestock animals **10** on the farm.

(48) The remote computer device in the monitoring system **200** may be referred to as a gateway server **210** that communicates with the network of monitoring devices **100**. The gateway server **210** manages secure gateway connections with and configurations of the monitoring devices **100**.

(49) The monitoring system **200** further includes a network server **220** that implements the LoRaWAN® protocol and is communicative with the gateway server **210**. The network server **220** is configured for validating the authenticity of the monitoring devices **100** and authenticating the data received from the monitoring devices **100** via the gateway server **210**. More specifically, the gateway server **210** receives the temperature data at predefined intervals from each respective monitoring device **100**, and transmits the temperature data to the network server **220**.

(50) The monitoring system **200** further includes an application server **230** communicative with the network server **220**. The application server **230** is configured for decrypting the data received from the network server **220** and encrypting the data transmitted to the network server **220**. More specifically, the gateway server **210** forwards the temperature data from the monitoring devices **100** to the network server **220**, which then authenticates the temperature data and forwards it to the application server **230**. The application server **230** then decrypts and processes the temperature data.

(51) The monitoring system **200** may further include an online dashboard or interface **240** for

viewing the processed temperature data. The online interface **240** may include various functionalities such as data analytics to analyse the temperature data and evaluate the health of the livestock animals **10**. Suitable machine learning algorithms may be implemented and trained with a temperature dataset to determine various health conditions of the livestock animals **10**, such as fever or onset of diseases.

(52) Some non-limiting examples of the gateway server **210**, network server **220**, and application server **230** include computers, laptops, mini-computers, mainframe computers, any non-transient and tangible machines that can execute a machine-readable code, cloud-based servers, distributed server networks, and a network of computer systems. As used herein, a server is a physical or cloud data processing system on which a server program runs. The server may be implemented in hardware or software, or a combination thereof. Additionally, the gateway server **210**, network server **220**, and application server **230** each includes a processor, a memory, and various other modules or components. The modules and components thereof are configured for performing various operations or steps and are configured as part of the processor. Such operations or steps are performed in response to non-transitory instructions operative or executed by the processor. The memory is used to store instructions and perhaps data which are read during program execution. The memory may be referred to in some contexts as computer-readable storage media and/or non-transitory computer-readable media. Non-transitory computer-readable media include all computer-readable media, with the sole exception being a transitory propagating signal per se.

(53) Each monitoring device **100** can thus monitor the body temperature of a respective livestock animal **10** remotely and passively, as the monitoring device **100** is orally administered into the livestock animal **10** and resides in the stomach **12** without being excreted. The size of the monitoring device **100** makes it suitable for oral administration into small young livestock animals **10**, even those as young as 2 weeks old. For ruminants, the body temperature may be referred to as the intraruminal temperature as it is measured inside the rumen. Notably, the intraruminal temperature is independent of external factors and cannot be manipulated from the outside, but it is usually 0.5° C. higher than the core body temperature due to the activity of heat-producing microbes in the rumen.

(54) The monitoring system **200** can be implemented with any LoRaWAN® network and the monitoring device **100** measures the body temperature and sends the temperature data to the gateway server **210**. The temperature data is subsequently processed and can be monitored on the online interface **240**. The monitoring device **100** obviates the need for manual measuring of body temperature, which is labour-intensive, and enables real-time remote monitoring of the body temperature.

(55) Monitoring of the body temperature is one of the most effective ways of evaluating the health of livestock animals **10**. Farms can more accurately assesses and predict good health of livestock animals **10** based on their body temperature. Real-time monitoring of the animal's health aids in the early detection of diseases. This allows the farm workers to quickly respond to any possible occurrence of diseases and treat the livestock animal **10** accordingly. Early treatment of livestock animals **10** reduces their mortality rate and thereby increase the livestock farm's productivity which dependent on their health. Labour costs for the farm can also be reduced as less manpower is needed to operate the monitoring system **200** and the monitoring devices **100**, as opposed to manual measurements using thermometers.

(56) The normal body temperature range for a livestock animal **10** may differ depending on various factors, such as species, gender, and age. For example, the normal body temperature may range from 38.3 to 38.8° C. for adult cattle, and from 38.5 to 39.5° C. for young calves. Whenever the livestock animal **10** consumes water, the temperature would drop a little. Thus, the temperature data can be monitored to determine whether the livestock animal **10** is consuming water at appropriate intervals. If the livestock animal **10** is found to have not been consuming water for an extended time period, farm workers may intervene by moving it to a water source or investigate for

suspicious health deficiency.

(57) If the livestock animal **10** is not consuming enough water, or if the temperature data shows the body temperature deviating from the normal range, the farm workers may take preventive measures to improve the health of the livestock animal **10**. For example, the farm workers can adequately prescribe the right intervention, such as hydrating the livestock animal **10** and feeding nutritious animal feed to ensure optimal health is maintained.

(58) If the temperature data shows the body temperature constantly above the normal range, then this could mean that the livestock animal **10** is febrile, which usually indicates an infection or disease. The health condition of the livestock animal **10** should be evaluated further, such as by determining whether the livestock animal **10** has other symptoms like lethargy and lack of appetite. Farm workers can then promptly prescribe the right resources to treat the livestock animal **10** before the health condition worsens, such as by administering medication (e.g. antibiotics), and/or health supplements (e.g. vitamins). The livestock animal **10** should also be isolated to prevent spread or outbreak of any infection or disease to other livestock animals **10**. This could help the farm save costs as any potential losses from such infection or disease would be limited to only one or a few livestock animals **10**.

(59) If the livestock animal **10** is female, elevations in the body temperature could indicate that the livestock animal is on heat, meaning that the female livestock animal **10** is in oestrus which is the receptive period of the sexual cycle. Suitable measures should then be taken for the female livestock animal **10** to mate with a male livestock animal **10**. Moreover, by deliberating pairing certain species of male and female livestock animals **10** for mating, the farm breed certain species of livestock and increase their population, especially those that are commercially valuable to the farm.

(60) As the monitoring device **100** can be used in young livestock animals **10**, monitoring their body temperature can reduce rate of premature deaths that could happen due to infection or disease, especially since young livestock animals **10** have weaker immunity. The young livestock animals **10** can continue to grow healthily into adults before they are processed, e.g. culled, for sale. Notably, adult livestock is often more valuable than young livestock as adult livestock is heavier and can yield more meat. With lower mortality rates, a greater number of healthy livestock animals **10** can be processed for sale, thus increasing revenue for the farm.

(61) Additionally, the monitoring system **200** implements LoRaWAN® protocol which provides for long range communication and a large network capacity. A large number of monitoring devices **100** can thus be connected to a single gateway server **210**. This significantly reduces the costs of deploying multiple gateway servers **210** and the manpower needed to operate and maintain them. Farms can manage their farm resources more appropriately by deploying the right resources in a timely manner, and as and when required. Thus, use of the monitoring devices **100** and the monitoring system **200** can help reduce operating costs, as well as increase productivity and revenue stream for the farms.

(62) In the foregoing detailed description, embodiments of the present disclosure in relation to a monitoring system and device for livestock are described with reference to the provided figures. The description of the various embodiments herein is not intended to call out or be limited only to specific or particular representations of the present disclosure, but merely to illustrate non-limiting examples of the present disclosure. The present disclosure serves to address at least one of the mentioned problems and issues associated with the prior art. Although only some embodiments of the present disclosure are disclosed herein, it will be apparent to a person having ordinary skill in the art in view of this disclosure that a variety of changes and/or modifications can be made to the disclosed embodiments without departing from the scope of the present disclosure. Therefore, the scope of the disclosure as well as the scope of the following claims is not limited to embodiments described herein.

Claims

1. A monitoring device for remotely monitoring health of a livestock animal, the monitoring device comprising: an elongated body within predefined dimensional limits that enable the monitoring device to be orally administrable into the livestock animal and to remain inside its stomach without being excreted; a set of electronic components housed within the elongated body, the electronic components comprising: a battery for powering the electronic components; a temperature sensor for measuring body temperature of the livestock animal; a computer processor for processing the measured temperature into temperature data; and a wireless communications component for transmitting the temperature data to a remote computer device for processing to monitor the health of the livestock animal; and a magnetic mechanism housed within the elongated body and configured for: activating the monitoring device in response to the magnetic mechanism detecting magnetic interference from an external magnetic device; and registering the monitoring device with the remote computer device to thereby pair the monitoring device with the remote computer device, said registration comprises transmitting, in response to said activation of the monitoring device, an activation data packet to the remote computer device for matching with predefined identification data of the monitoring device, wherein the magnetic mechanism is configured for resetting the monitoring device to an inactive mode in order to reattempt registration with the remote computer device; and wherein the monitoring device is configured to disable the magnetic mechanism after said registration to prevent resetting of the monitoring device, thereby permanently pairing the monitoring device with the remote computer device.
2. The monitoring device according to claim 1, wherein the elongated body has an overall width of up to 25 mm, wherein the elongated body has an overall length of up to 100 mm, and/or wherein the monitoring device has an overall weight from 85 g to 95 g.
3. The monitoring device according to claim 1, wherein the elongated body is tubular and comprises a rounded front portion to facilitate oral administration of the monitoring device.
4. The monitoring device according to claim 1, wherein the monitoring device is front weighted to facilitate oral administration of the monitoring device.
5. The monitoring device according to claim 1, wherein the electronic components further comprise a memory component connected to the computer processor for storing the temperature data.
6. The monitoring device according to claim 1, wherein the wireless communications component is configured for transmission using a LoRa wireless communication protocol.
7. The monitoring device according to claim 1, wherein the computer processor is configured for controlling the temperature sensor and wireless communications component to perform said measuring and transmitting at predefined intervals.
8. The monitoring device according to claim 1, wherein said reset is performed in response to the magnetic mechanism detecting magnetic interference from the external magnetic device.
9. The monitoring device according to claim 8, further comprising a lighting element housed within the elongated body and configured for visual indication of activation of the monitoring device.
10. The monitoring device according to claim 1, further comprising a data matrix code for retrieving the predefined identification data of the monitoring device.
11. The monitoring device according to claim 10, further comprising a detachable label attached on the elongated body, wherein the data matrix code is visible on the label.
12. The monitoring device according to claim 1, further comprising a human-readable identifier of the monitoring device integrally formed on the elongated body.
13. The monitoring device according to claim 1, wherein the elongated body comprises a thermal epoxy material, and/or wherein the elongated body is coated with a food grade polypropylene material.
14. The monitoring device according to claim 1, wherein the electronic components further

comprise a RFID transponder to enable electronic identification of the livestock animal.

15. A monitoring system for remotely monitoring health of livestock animals, the monitoring system comprising: a set of monitoring devices; a remote computer device communicative with the monitoring devices; and each monitoring device residing in a stomach of a respective livestock animal, the monitoring device comprising: an elongated body within predefined dimensional limits that enable the monitoring device to be orally administrable into the livestock animal to remain inside its stomach without being excreted; a set of electronic components housed within the elongated body, the electronic components comprising: a battery for powering the electronic components; a temperature sensor for measuring temperature of the livestock animal; a computer processor for processing the measured temperatures into temperature data; and a wireless communications component for transmitting the temperature data to the remote computer device; and a magnetic mechanism housed within the elongated body and configured for: activating the monitoring device in response to the magnetic mechanism detecting magnetic interference from an external magnetic device; and registering the monitoring device with the remote computer device, said registration comprises transmitting, in response to said activation of the monitoring device, an activation data packet to the remote computer device for matching with predefined identification data of the monitoring device, wherein the magnetic mechanism is configured for resetting the monitoring device to an inactive mode in order to reattempt registration with the remote computer device; and wherein the monitoring device is configured to disable the magnetic mechanism after said registration to prevent resetting of the monitoring device, thereby permanently pairing the monitoring device with the remote computer device, wherein the remote computer device is configured for processing the temperature data to monitor the health of the livestock animals.

16. The monitoring system according to claim 15, wherein communication between the monitoring devices and the remote computer device is based on the LoRa wireless communication protocol.

17. The monitoring system according to claim 15, wherein said reset is performed in response to the magnetic mechanism detecting magnetic interference from the external magnetic device.
